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Contents

1 Biographical Narrative 2
2 Professional Preparation 3
3 Appointments 3
4 Honors 4
5 Publications 5
6 Presentations 24
7 Software Products 32
8 Patents 32
9 Professional Activities and Service 33
10 Funding 38
11 Teaching and Student Guidance 41
12 Personal 42
13 References 43

1 Biographical Narrative

Jeffrey Vetter, Ph.D., is a Corporate Fellow at Oak Ridge National Laboratory (ORNL). At ORNL, he is currently the Section Head for Advanced Computer Systems Research and the founding director of the Experimental Computing Laboratory (ExCL). Previously, Vetter was the founding group leader of the Future Technologies Group in the Computer Science and Mathematics Division from 2003 until 2020. From 2006 through 2016, Vetter held a Joint position at Georgia Institute of Technology. At GT from 2009 to 2015, Vetter was the Principal Investigator of the NSF Track 2D Experimental Computing XSEDE Facility, named Keeneland, for large scale heterogeneous computing using graphics processors, and the Director of the NVIDIA CUDA Center of Excellence. From 2016 until 2020, Vetter held a joint appointment at the Electrical Engineering and Computer Science Department of the University of Tennessee-Knoxville.

Vetter earned his Ph.D. in Computer Science from the Georgia Institute of Technology. He joined ORNL in 2003, after stints as a computer scientist and project leader at Lawrence Livermore National Laboratory, and postdoctoral researcher at the University of Illinois at Urbana-Champaign. The coherent thread through his research is developing rich architectures and software systems that solve important, real-world high-performance computing problems. He has been investigating the effectiveness of radically diverse architectures, including deep memory hierarchies and heterogeneous processors such as GPUs, SoCs, and field-programmable gate arrays (FPGAs), for important mission applications; in fact, he helped to coin the term *Extreme Heterogeneity*. His recent books, entitled "Contemporary High Performance Computing: From Petascale toward Exascale (Vols. 1 – 3)," survey the international landscape of HPC.

Vetter is a Fellow of the IEEE and AAAS, and a Distinguished Scientist Member of the ACM. In 2018, Vetter was awarded the ORNL Director's Award for Outstanding Individual Accomplishment in Science and Technology. As part of an interdisciplinary team from Georgia Tech, NYU, and ORNL, Vetter was awarded the 2010 Gordon Bell Prize. In 2020, along with a large team from IBM and LLNL, Vetter received the SC20 Test of Time Award for the paper from SC02, entitled "An Overview of the Blue Gene/L Supercomputer." Also, his work has won awards at major venues: Best Paper Awards at the International Parallel and Distributed Processing Symposium (IPDPS), EuroPar, and the 2018 AsHES Workshop, Best Student Paper Finalist at SC14, Best Paper Finalist at the IEEE HPEC Conference, and Best Presentation at EASC 2015. In 2015, Vetter served as the Technical Program Chair of SC15 (SC15 Breaks Exhibits and Attendance Records While in Austin).

Vetter is a frequent presenter as a keynote, plenary, and invited speaker at conferences (e.g., SC, ISC, HPDC, PASC, SIAM PP), major international HPC organizations (e.g., RIKEN R-CCS, Swiss National Supercomputing Center, Barcelona Supercomputing Center, IBM TJ Watson Research Center, Argonne Training Program on Extreme-Scale Computing, Rocky Mountain Advanced Computing Consortium Symposium, Human Brain Project at EPFL, Schloss Dagstuhl, Julich Supercomputing Centre (JSC), Lawrence Berkeley National Laboratory, Argonne National Laboratory), universities (e.g., Tokyo Institute of Technology, Purdue, University of Texas at Austin, University of Illinois at Urbana Champaign, University of Michigan-Ann Arbor) and government organizations (e.g., NITRD Future Computing Community of Interest Meeting, DOE Centers of Excellence Performance Portability Meeting, DOE Extreme Heterogeneity Workshop).

Recent sponsors of his team's work include the Department of Energy Office of Science (SC), the Defense Advanced Research Projects Agency (DARPA), the National Science Foundation (NSF), and Oak Ridge National Laboratory. Currently, Vetter serves as a level 3 manager and software project PI in the DOE Exascale Computing Project (ECP), a PI for the Domain-Specific System on a Chip (DSSoC) program in the DAPRA Electronics Resurgence Initiative, the PI of the ASCR Sawtooth project to investigate solutions to challenges in emerging memory and storage systems, and Platform Readiness Thrust Lead in the DOE SciDAC RAPIDS Institute, in addition to other roles.

2 Professional Preparation

1. Postdoctoral Research Associate. University of Illinois, Urbana-Champaign. 1998-9.
2. Ph.D. in Computer Science. Georgia Institute of Technology. Atlanta. 1998.
3. M.S. in Computer Science. Georgia Institute of Technology. Atlanta. 1994.
4. B.S. in Electrical/Computer Engineering w/ Honors. Georgia Institute of Technology. Atlanta. 1989.

3 Appointments

1. Oak Ridge National Laboratory.
   (a) Section Head for Advanced Computing Research Systems. October 2020 - present.
   (b) Corporate Fellow. August 2020 - present.
   (c) Founding Director of Experimental Computing Laboratory (ExCL). 2004 - present.
   (d) Distinguished R&D Staff Member and Group Leader. November 2009 - August 2020.
   (e) Founding Group Leader of ORNL Future Technologies Group. 2004 - 2020.
   (f) Senior R&D Staff Member. November 2003 - November 2009
2. University of Tennessee - Knoxville, Department of Electrical Engineering and Computer Science.
   (a) Joint Professor. 2016 - 2020
3. Georgia Institute of Technology, College of Computing.
   (a) Joint Professor. 2006 - 2016.
   (b) Principal Investigator and Director, NVIDIA CUDA Center of Excellence. 2010 - 2016.
   (c) Principal Investigator and Project Director, NSF Keeneland Project. 2009 - 2015.
   (d) Adjunct Professor. May 2005 - October 2006.
4. Lawrence Livermore National Laboratory. Computer Scientist and Project Leader. 1999 - 2003.
5. Los Alamos National Laboratory. Graduate research intern. Summers of 1996, 1997.
6. Intel Corporation. Graduate research intern. Sept 1991 - Dec 1992, Summer 1993.
7. Hewlett Packard Corporation (now Agilent Technologies). Technical staff member. 1989 - 1991.
8. Georgia Institute of Technology, Undergraduate research assistant. Experiment control system for analysis of CCD semiconductor materials. University semesters during 1987-1989.

4 Honors

1. Institute of Electrical and Electronic Engineers (IEEE)
   - **Senior** member, 2012.
   - **Fellow**, 2016.
2. Association of Computing Machinery (ACM)
   - **Distinguished Scientist** member, 2012.
3. American Association for the Advancement of Science (AAAS)
   - **Fellow**, 2020.
4. ORNL/UT-Battelle **Director's Award for Outstanding Individual Accomplishment in Science and Technology**, 2018. [link]
5. ORNL/UT-Battelle Award for **Distinguished Researcher** (1 of 2), 2018. [link]
6. **SC20 Test of Time Award**: "An Overview of the BlueGene/L Supercomputer." N. Adiga, G. Almasi et al (2002). 2002 ACM/IEEE Conference on Supercomputing. [link]
7. **ACM/IEEE Gordon Bell Prize**: "Petascale Direct Numerical Simulation of Blood Flow on 200K Cores and Heterogeneous Architectures." A. Rahimian, I. Lashuk, S. Veerapaneni, A. Chandramowlishwaran, D. Malhotra, L. Moon, R. Sampath, A. Shringarpure, J. Vetter, R. Vuduc, D. Zorin and G. Biros (2010). 2010 ACM/IEEE International Conference for High Performance Computing, Networking, Storage and Analysis (SC10).
8. **Best Paper**: "NVIDIA Tensor Core Programmability, Performance & Precision," S. Markidis, S. Wei Der Chien, E. Laure, I.B. Peng, and J.S. Vetter, Proc. Eighth International Workshop on Accelerators and Hybrid Exascale Systems (AsHES) 2018.
9. **Best Student Paper Finalist**: L. Yu, D. Li, S. Mittal, J.S. Vetter, "Quantitatively Modeling Application Resiliency with the Data Vulnerability Factor," in SC14: ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis. New Orleans, LA, 2014
10. **Best Paper Finalist**: M.E. Belviranli, S. Lee, and J.S. Vetter, "Designing Algorithms for the EMU Migrating-threads-based Architecture," in IEEE HPEC, 2018.
11. **Award Paper**: S.R. Alam and J.S. Vetter, Hierarchical Model Validation of Symbolic Performance Models of Scientific Applications, Proc. Euro-Par 2006.
12. **Award Paper**: J.S. Vetter and F. Mueller (2002). Communication Characteristics of Large-Scale Scientific Applications for Contemporary Cluster Architectures. International Parallel and Distributed Processing Symp. (IPDPS).
13. **Best Presentation Award**: Exascale Applications and Software Conference (EASC) for "Exploring Emerging Technologies in the Extreme Scale HPC Co-Design Space with Holistic Performance Modeling," 2015.
14. **Most Distinguished Software**: ORNL Computer Science and Mathematics Division Award for Scalable Heterogeneous Computing Benchmark suite, 2013.
15. NASA Graduate Research Fellowship, 1993-7.
16. LLNL Computation Directorate Award for ASCI PSE milestone contributions, 2001.
17. LLNL Computation Directorate Award for technical contributions on ASCI Purple and BlueGene/L procurement, 2002.
18. LLNL Computation Directorate Award for ASCI performance modeling and analysis contributions, 2003.

5 Publications

Google Scholar citations are available at http://scholar.google.com/citations?hl=en&user=pWnx6r4AAAAJ.

5.1 Books

1. J. S. Vetter, editor. *Contemporary High Performance Computing: From Petascale Toward Exascale*, volume 3 of *CRC Computational Science Series*. Taylor and Francis, Boca Raton, 1 edition, 2018

2. J. S. Vetter, editor. *Contemporary High Performance Computing: From Petascale Toward Exascale*, volume 2. Taylor and Francis, Boca Raton, 1 edition, 2015a

3. J. S. Vetter, editor. *Contemporary High Performance Computing: From Petascale Toward Exascale*, volume 1 of *CRC Computational Science Series*. Taylor and Francis, Boca Raton, 1 edition, 2013a

5.2 Book Chapter

1. J. S. Vetter. Terascale to petascale: The past 17 years in high performance computing. In *Contemporary High Performance Computing*, volume 2 of *Chapman & Hall/CRC Computational Science*, pages 1–5. Taylor and Francis, Boca Raton, 1 edition, 2015b. doi:doi:10.1201/b18256-2

2. J. S. Vetter, S. Lee, D. Li, G. Marin, C. McCurdy, J. Meredith, P. C. Roth, and K. Spafford. Quantifying architectural requirements of contemporary extreme-scale scientific applications. In Stephen A. Jarvis, Steven A. Wright, and Simon D. Hammond, editors, *High Performance Computing Systems. Performance Modeling, Benchmarking and Simulation: 4th International Workshop, PMBS 2013, Denver, CO, USA, November 18, 2013. Revised Selected Papers*, pages 3–24. Springer International Publishing, Cham, 2014. doi:10.1007/978-3-319-10214-6_1

3. J. S. Vetter, R. Glassbrook, K. Schwan, S. Yalamanchili, M. Horton, A. Gavrilovska, M. Slawinska, J. Dongarra, J. Meredith, P. C. Roth, K. Spafford, S. Tomov, and J. Wynkoop. Keeneland: Computational science using heterogeneous GPU computing. In Jeffrey S. Vetter, editor, *Contemporary High Performance Computing: From Petascale Toward Exascale*, volume 1 of *CRC Computational Science Series*, page 730. Taylor and Francis, Boca Raton, 1 edition, 2013a

4. J. S. Vetter. Contemporary high performance computing: an introduction. In Jeffrey S. Vetter, editor, *Contemporary High Performance Computing: From Petascale Toward Exascale*, volume 1 of *CRC Computational Science Series*, page 730. Taylor and Francis, Boca Raton, 1 edition, 2013b

5. J. S. Vetter, V. Tipparaju, W. Yu, and P. C. Roth. HPC interconnection networks: The key to exascale computing. In *Advances in Parallel Computing: High Speed and Large Scale Scientific Computing*, volume 18, pages 95–106. 2009

6. J. Dongarra, R. Graybill, W. Harrod, R. Lucas, E. Lusk, P. Luszczek, J. McMahon, A. Snavely, J. Vetter, K. Yelick, S. Alam, R. Campbell, L. Carrington, T.-Y. Chen, O. Khalili, J. Meredith, and M. Tikir. DARPA's HPCS program: History, models, tools, languages. In V. Zelkowitz Marvin, editor, *Advances in Computers*, volume Volume 72, pages 1–100. Elsevier, 2008. doi:10.1016/s0065-2458(08)00001-6

7. N. Bhatia and J. S. Vetter. Virtual cluster management with xen. In *Euro-Par 2007 Workshops: Parallel Processing*, volume 4854/2008 of *Lecture Notes in Computer Science*, pages 185–194. Springer, Berlin / Heidelberg, 2007

8. L. Derose, M. Pantano, J. S. Vetter, and D. A. Reed. Performance issues in parallel processing systems. In Gunter Haring, Christoph Lindemann, and Martin Reiser, editors, *Performance Evaluation - Origins and Directions*, page 527. Springer-Verlag, 2000

9. G. Eisenhauer, W. Gu, T. Kindler, K. Schwan, D. Silva, and J. S. Vetter. Opportunities and tools for highly interactive distributed and parallel computing. In Margaret Simmons, Ann Hayes, Jeffrey Brown, and Dan Reed, editors, *Debugging and Performance Tuning for Parallel Computing Systems*, pages 245–77. IEEE Computer Society Press, 1996

5.3 Journal Article

1. G. Papadimitriou, C. Wang, K. Vahi, R. F.d. Silva, A. Mandal, Z. Liu, R. Mayani, M. Rynge, M. Kiran, V. E. Lynch, R. Kettimuthu, E. Deelman, J. S. Vetter, and I. T. Foster. End-to-end online performance data capture and analysis for scientific workflows. *Future Gener. Comput. Syst.*, 117:387–400, 2021. doi:10.1016/j.future.2020.11.024

2. J. Lambert, S. Lee, J. S. Vetter, and A. D. Malony. Optimization with the openACC-to-FPGA framework on the arria 10 and stratix 10 FPGAs. *Parallel Comput.*, 104-105:102784, 2021. doi:10.1016/j.parco.2021.102784

3. G. J. Herschlag, S. Lee, J. S. Vetter, and A. Randles. Analysis of GPU data access patterns on complex geometries for the D3Q19 lattice boltzmann algorithm. *IEEE Trans. Parallel Distributed Syst.*, 32(10):2400–2414, 2021. doi:10.1109/TPDS.2021.3061895

4. R. Tsunashima, R. Kobayashi, N. Fujita, T. Boku, S. Lee, J. S. Vetter, H. Murai, M. Nakao, and M. Sato. OpenACC unified programming environment for GPU and FPGA multi-hybrid acceleration. *Miguel Areias Inês Dutra Ricardo Rocha (Eds.)*, page 114, 2020

5. S. Lee, J. Gounley, A. Randles, and J. S. Vetter. Performance portability study for massively parallel computational fluid dynamics application on scalable heterogeneous architectures. *Journal of Parallel and Distributed Computing*, 129:1–13, 2019. doi:https://doi.org/10.1016/j.jpdc.2019.02.005

6. J. Kim and J. S. Vetter. Implementing efficient data compression and encryption in a persistent key-value store for HPC. *The International Journal of High Performance Computing Applications*, 33(6): 1098–1112, 2019. doi:10.1177/1094342019847264

7. S. W. D. Chien, S. Markidis, V. Olshevsky, Y. Bulatov, E. Laure, and J. S. Vetter. Tensorflow doing HPC. *CoRR*, abs/1903.04364, 2019

8. M. Umar, S. V. Moore, J. S. Meredith, J. S. Vetter, and K. W. Cameron. Aspen-based performance and energy modeling frameworks. *Journal of Parallel and Distributed Computing*, 120:222–236, 2018a. doi:https://doi.org/10.1016/j.jpdc.2017.11.005

9. I. B. Peng, R. Gioiosa, G. Kestor, J. S. Vetter, P. Cicotti, E. Laure, and S. Markidis. Characterizing the performance benefit of hybrid memory system for HPC applications. *Parallel Computing*, 76:57–69, 2018a. doi:https://doi.org/10.1016/j.parco.2018.04.007

10. J. S. Vetter, E. P. DeBenedictis, and T. M. Conte. Architectures for the post-moore era. *IEEE Micro*, 37(4):6–8, 2017a

11. M. Umar, S. V. Moore, J. S. Meredith, J. S. Vetter, and K. W. Cameron. Aspen-based performance and energy modeling frameworks. *Journal of Parallel and Distributed Computing*, 2017. doi:https://doi.org/10.1016/j.jpdc.2017.11.005

12. S. Mittal, R. Wang, and J. S. Vetter. DESTINY: A comprehensive tool with 3D and multi-level cell memory modeling capability. *Journal of Low Power Electronics and Applications*, 7(3):23–35, 2017a

13. E. Deelman, C. Carothers, A. Mandal, B. Tierney, J. S. Vetter, I. Baldin, C. Castillo, G. Juve, D. Krol, V. Lynch, B. Mayer, J. Meredith, T. Proffen, P. Ruth, and R. F. da Silva. PANORAMA: An approach to performance modeling and diagnosis of extreme-scale workflows. *International Journal of High Performance Computing Applications*, 31(1):4–18, 2017a. doi:10.1177/1094342015594515

14. E. Deelman, T. Peterka, I. Altintas, C. D. Carothers, K. K.v. Dam, K. Moreland, M. Parashar, L. Ramakrishnan, M. Taufer, and J. Vetter. The future of scientific workflows. *The International Journal of High Performance Computing Applications*, 0(0):1094342017704893, 2017b. doi:10.1177/1094342017704893

15. S. Mittal and J. Vetter. A technique for improving lifetime of non-volatile caches using write-minimization. *Journal of Low Power Electronics and Applications*, 6(1):1, 2016a

16. S. Mittal and J. S. Vetter. A survey of software techniques for using non-volatile memories for storage and main memory systems. *IEEE Transactions on Parallel and Distributed Systems*, 27(5):1537–1550, 2016b. doi:10.1109/TPDS.2015.2442980

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18. S. Mittal and J. S. Vetter. A survey of techniques for modeling and improving reliability of computing systems. *IEEE Transactions on Parallel and Distributed Systems*, 27(4):1226–1238, 2016d. doi:10.1109/Tpds.2015.2426179

19. S. Mittal and J. S. Vetter. A survey of architectural approaches for data compression in cache and main memory systems. *IEEE Transactions on Parallel and Distributed Systems*, 27(5):1524–1536, 2016e. doi:10.1109/Tpds.2015.2435788

20. S. Mittal and J. S. Vetter. Equalwrites: Reducing intra-set write variations for enhancing lifetime of non-volatile caches. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 24(1):103–114, 2016f. doi:10.1109/Tvlsi.2015.2389113

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22. J. S. Vetter and S. Mittal. Opportunities for nonvolatile memory systems in extreme-scale high performance computing. *Computing in Science and Engineering*, 17(2):73–82, 2015. doi:10.1109/mcse.2015.4

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24. A. Sabne, P. Sakdhnagool, S. Lee, and J. S. Vetter. Understanding portability of a high-level programming model on contemporary heterogeneous architectures. *Micro, IEEE*, 35(4):48–58, 2015. doi:10.1109/MM.2015.73

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26. S. Mittal, J. S. Vetter, and D. Li. A survey of architectural approaches for managing embedded DRAM and non-volatile on-chip caches. *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 26 (6):1524–1537, 2015. doi:10.1109/tpds.2014.2324563

27. S. Mittal and J. S. Vetter. A survey of methods for analyzing and improving GPU energy efficiency. *ACM Computing Surveys*, 47(2):19:23, 2015b. doi:10.1145/2636342

28. S. Mittal and J. S. Vetter. A survey of CPU-GPU heterogeneous computing techniques. *ACM Computing Surveys*, 47(4):1–35, 2015c. doi:10.1145/2788396

29. S. Mittal and J. Vetter. A survey of techniques for modeling and improving reliability of computing systems. *IEEE Transactions on Parallel and Distributed Systems (TPDS)*, 27(4):1226–1238, 2015d. doi:10.1109/tpds.2015.2426179

30. S. Mittal and J. S. Vetter. Equalwrites: Reducing intra-set write variations for enhancing lifetime of non-volatile caches. *IEEE Transactions on VLSI Systems*, 24(1):103–114, 2015e. doi:10.1109/tvlsi.2015.2389113

31. S. Mittal and J. S. Vetter. AYUSH: A technique for extending lifetime of SRAM-NVM hybrid caches. *IEEE Computer Architecture Letters (CAL)*, 14(2):115–118, 2015f. doi:10.1109/lca.2014.2355193

32. E. Deelman, C. Carothers, A. Mandal, B. Tierney, J. S. Vetter, I. Baldin, C. Castillo, G. Juve, D. Krol, V. Lynch, B. Mayer, J. Meredith, T. Proffen, P. Ruth, and R. Ferreira da Silva. PANORAMA: An approach to performance modeling and diagnosis of extreme-scale workflows. *International Journal of High Performance Computing Applications*, 2015. doi:10.1177/1094342015594515

33. L. Yu, D. Li, S. Mittal, and J. S. Vetter. Quantitatively modeling application resiliency with the data vulnerability factor. *ACM/IEEE International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, 2014. doi:10.1109/SC.2014.62

34. Y. Tian, C. Xu, W. Yu, J. Vetter, S. Klasky, H. Liu, and S. Biaz. neCODEC: nearline data compression for scientific applications. *Cluster Computing*, pages 1–12, 2013. doi:10.1007/s10586-013-0265-8

35. K. Spafford, J. S. Vetter, T. Benson, and M. Parker. Modeling synthetic aperture radar computation with aspen. *International Journal of High Performance Computing Applications*, 27(3):255–262, 2013. doi:10.1177/1094342013488262

36. A. Danalis, P. Luszczek, G. Marin, J. S. Vetter, and J. Dongarra. Blackjackbench: Portable hardware characterization with automated results' analysis. *The Computer Journal*, 2013

37. W. Yu, X. Que, V. Tipparaju, and J. S. Vetter. HiCOO: Hierarchical cooperation for scalable communication in global address space programming models on cray XT systems. *Journal of Parallel and Distributed Computing*, 2012. doi:10.1016/j.jpdc.2012.01.022

38. V. Tipparaju, E. Apra, W. Yu, X. Que, and J. Vetter. Runtime techniques to enable a highly-scalable global address space model for petascale computing. *International Journal of Parallel Programming*, 40(6):1–23, 2012. doi:10.1007/s10766-012-0214-9

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40. J. S. Vetter, R. Glassbrook, J. Dongarra, K. Schwan, B. Loftis, S. McNally, J. Meredith, J. Rogers, P. Roth, K. Spafford, and S. Yalamanchili. Keeneland: Bringing heterogeneous GPU computing to the computational science community. *IEEE Computing in Science and Engineering*, 13(5):90–95, 2011. doi:10.1109/MCSE.2011.83

41. J. S. Meredith, P. C. Roth, K. L. Spafford, and J. S. Vetter. Performance implications of nonuniform device topologies in scalable heterogeneous architectures. *IEEE Micro*, 31(5):66–75, 2011. doi:10.1109/MM.2011.79

42. J. Dongarra, P. Beckman, T. Moore, P. Aerts, G. Aloisio, J.-C. Andre, D. Barkai, J.-Y. Berthou, T. Boku, B. Braunschweig, F. Cappello, B. Chapman, C. Xuebin, A. Choudhary, S. Dosanjh, T. Dunning, S. Fiore, A. Geist, B. Gropp, R. Harrison, M. Hereld, M. Heroux, A. Hoisie, K. Hotta, J. Zhong, Y. Ishikawa, F. Johnson, S. Kale, R. Kenway, D. Keyes, B. Kramer, J. Labarta, A. Lichnewsky, T. Lippert, B. Lucas, B. Maccabe, S. Matsuoka, P. Messina, P. Michielse, B. Mohr, M. S. Mueller, W. E. Nagel, H. Nakashima, M. E. Papka, D. Reed, M. Sato, E. Seidel, J. Shalf, D. Skinner, M. Snir, T. Sterling, R. Stevens, F. Streitz, B. Sugar, S. Sumimoto, W. Tang, J. Taylor, R. Thakur, A. Trefethen, M. Valero, A. van der Steen, J. Vetter, P. Williams, R. Wisniewski, and K. Yelick. The international exascale software project roadmap. *International Journal of High Performance Computing Applications*, 25(1): 3–60, 2011. doi:10.1177/1094342010391989

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team. *Journal of Physics: Conference Series (Proceedings of SciDAC 2009, San Diego, CA, July 14-18, 2009)*, 2009

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22. J. S. Vetter. Experiences using computational steering on existing scientific applications. Technical report, 1999

23. W. Ribarsky, J. Tumblin, G. Newton, R. Nowicki, and J. S. Vetter. GLYPHMAKER: An interactive, programmerless approach for customizing, exploring, and analyzing visual data representatives. Technical Report GIT-GVU-93-26, Graphics, Visualization, and Usability Center, Georgia Tech, 1994

5.6 PhD Thesis

1. J. S. Vetter. *Techniques and optimizations for high performance computational steering*. Ph.d., 1998

6 Presentations

Vetter was the speaker for all of the presentations listed in this section.

6.1 Keynote

1. *Preparing for Extreme Heterogeneity in High Performance Computing*, 33rd Workshop on Languages and Compilers for Parallel Computing (LCPC) , Stony Brook University, NY , 2020.
2. *Preparing for Extreme Heterogeneity in High Performance Computing*, RIKEN R-CCS, Kobe, 2020.
3. *Preparing for Extreme Heterogeneity in HPC*, Barcelona Supercomputing Center, Barcelona, 2019.
4. *Preparing for Extreme Heterogeneity in High Performance Computing*, Rocky Mountain Advanced Computing Consortium Symposium , Boulder, 2019.
5. *Preparing for Extreme Heterogeneity in HPC*, PPAM 2019, Bialystok, Poland, 2019.
6. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , Exascale Application and Software Conference (EASC), Edinburgh, 2018.
7. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , 12th International Conference on Parallel Processing and Applications, Lublin, 2017.
8. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , ExaComm: Third International Workshop on Communication Architectures for HPC, Big Data, Deep Learning and Clouds at Extreme Scale, Frankfurt, 2017.
9. *Preparing for Supercomputing's Sixth Wave*, ACM Symposium on High-Performance Parallel and Distributed Computing (HPDC), Kyoto, 2016.
10. *Exploring Emerging Technologies in the HPC Co-design Space*, 2nd International HPC Forum, Tianjin, 2015.
11. *Exploring Emerging Technologies in the HPC Co-Design Space*, 11th International Conference on Parallel Processing and Applications, Krakow, 2015.
12. *Exploring Emerging Technologies in the HPC Co-design Space*, Workshop on Accelerator Programming using Directives (WACCPD), New Orleans, 2014.
13. *Exploring Emerging Technologies in the HPC Co-design Space*, AsHES Workshop (in conjunction with IPDPS), Phoenix, 2014.
14. *Exploring Emerging Technologies in the HPC Co-design Space*, International Conference on Energy-Aware High Performance Computing, Dresden, Germany, 2014.
15. *Exploring Emerging Technologies in the HPC Co-design Space*, FRCRC High Performance Computing Symposium, Laramie, WY, 2013.
16. *On the Road to Exascale: Lessons from Contemporary Scalable GPU Systems*, ATIP/A*CRC/NSF Workshop on Accelerator Technologies for High-Performance Computing, Singapore, 2012.
17. *Emerging Architectures on the Pathway to Exascale Computing*, HP-Cast, Hamburg, 2011.

6.2 Plenary

1. *The Coming Age of Extreme Heterogeneity*, ATPESC: Argonne Training Program on Extreme-Scale Computing , Chicago, 2019. [more]
2. *Architecting and Programming Future Extremely Heterogeneous Systems*, MODSIM 2019, Seattle, 2019.

3. *Preparing for Supercomputing's Sixth Wave*, CHPC National Meeting 2016, East London, South Africa, 2016.

4. *Exploring Emerging Technologies in the HPC Co-design Space*, 10th International Conf Parallel Processing and Applied Mathematics, Warsaw, 2013.

5. *Automated Exploration of the HPC Codesign Space with Aspen*, 1st International Workshop on Strategic Development of High Performance Computers, Tsukuba, Japan, 2013.

6. *Automated Exploration of the HPC Codesign Space*, International HPC Forum (IHPCF), Changsha, China, 2013.

7. *Emerging Technologies on the Pathway to Exascale*, PRACE Future Technologies Workshop, Daresbury, 2012.

8. *The Critical Role of Software Infrastructure in Future Exascale Systems*, 40th SPEEDUP Workshop, Basel, Switzerland, 2012.

9. *On the path to Exascale: Deploying an Emerging HPC Architecture*, DOE Exascale Tools Workshop, Annapolis, 2011.

10. *The Critical Role of Software Infrastructure in Future Exascale Systems*, NVIDIA GTC Asia, Beijing, 2011.

11. *Toward Exascale Computational Science: the Role of Emerging Technologies*, SC10 Broader Engagement Workshop, New Orleans, 2010.

12. *Toward Exascale Computational Science with Graphics Processors*, Cray Asia Technical Workshop, Tokyo, 2009.

13. *Pioneering Science Applications and Breakthroughs on Jaguar*, International Supercomputing Conference, Hamburg, Germany, 2009.

14. *Application Accelerators: Deus ex machina?*, High Performance Embedded Computing Workshop, Lexington, MA, 2009.

15. *Toward Exascale Computational Science with Graphics Processors*, NVIDIA GPU Technical Conference, San Jose, 2009.

16. *Embracing Uncertainty: Performance analysis and prediction on next-generation computer architectures*, DOE SciDAC Conference, San Francisco, 2005.

6.3 Invited

1. *A Perspective on AI in Computational Science and High Performance Computing*, MLSYS Workshop on Software-Hardware Codesign for ML Workloads, Austin, 2020.

2. *Preparing for Extreme Heterogeneity in High Performance Computing*, Australian National University, Canberra, 2020.

3. *Preparing for Extreme Heterogeneity in High Performance Computing*, Multicore World 2020, Wellington, 2020.

4. *Architecting and Programming Heterogeneous Memory Systems*, PASC 19, Zurich, 2019.

5. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , Multicore World 2019, Wellington, 2019.

6. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , University of Queensland, Brisbane, 2019.

7. *Preparing for Extreme Heterogeneity in HPC*, ASCR Applied Math PI Meeting, Rockville, MD, 2019.

8. *Preparing for Extreme Heterogeneity in HPC*, NITRD HEC IA Working Group, Washington, DC, 2019.
9. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing*, ISC, Frankfurt, 2019.
10. *Working toward performance portability for FPGAs in High Performance Computing*, 3rd International Workshop on FPGA for HPC (IWFH), Tokyo, 2018.
11. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , Queens University, Belfast, 2018.
12. *DRAGON: Efficiently Increasing GPU Memory Capacity with NVM* , 5th DOE/MEXT Workshop, Tokyo, 2018.
13. *Preparing for Extreme Heterogeneity in HPC*, Symposium on Materials Genome Initiative at Exascale, Spetses, Greece, 2018.
14. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , KTH Royal Institute of Technology, Stockholm, 2018.
15. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , TU Darmstadt, Darmstadt, 2018.
16. *Preparing for Extreme Heterogeneity in HPC*, CCDSC, Lyon, France, 2018.
17. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , 2017 High Performance Computing (HPC) and Data Analytics Workshop, Hanover, MD, 2017.
18. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , ICM, Warsaw, 2017.
19. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , National Supercomputing Center in Guangzhou, Guangzhou, 2017.
20. *Exploring Emerging Technologies in the HPC Co-Design Space*, SPPEXA Symposium, Munich, 2016.
21. *Performance Portability for Extreme Scale High Performance Reconfigurable Computing* , Workshop on FPGAs for Scientific Simulation and Data Analytics, NCSA, 2016.
22. *Performance Portability for Extreme Scale High Performance Computing* , Workshop on Clusters, Clouds, and Data for Scientific Computing, Lyon, 2016.
23. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing* , CHPC National Meeting 2016, East London, South Africa, 2016.
24. *Managing the Emerging Complexity Wall of Extreme Scale High Performance Computing*, University of Tennessee, Knoxville, Knoxville, 2016.
25. *Exploring Emerging Memory Technologies in Extreme Scale High Performance Computing*, Forschungszentrum Julich GmbH, Julich Supercomputing Centre (JSC), Julich, 2016.
26. *Managing the Memory Hierarchy*, SOS20, Asheville, 2016.
27. *Using Application-Specific Performance Models to Inform Dynamic Scheduling*, 11th Scheduling for Large Scale Systems Workshop, Nashville, 2016.
28. *Understanding Portability of a High-Level Programming Model on Diverse HPC Architectures*, ADAC Workshop Switzerland 2016, Lugano, 2016.
29. *Exploring Emerging Technologies for Extreme Scale High Performance Computing*, Ecole Polytechnique Federale de Lausanne (EPFL), Geneva, 2016.

30. *Performance Portability for Extreme Scale High Performance Computing* , International Workshop on HPC Architecture, Software, and Application at an Extreme Scale @ National Supercomputing Center in Wuxi, Wuxi, 2016.

31. *Performance Portability for Extreme Scale High Performance Computing* , National Supercomputing Center in Guangzhou, Guangzhou, 2016.

32. *Using Application-Specific Performance Models to Inform Dynamic Scheduling*, 6th International Workshop on Runtime and Operating Systems for Supercomputers (ROSS), Kyoto, 2016.

33. *Quantitatively Modeling Application Resilience with the Data Vulnerability Factor*, Salishan Conference on High-speed Computing, Gleneden Beach, OR, 2015.

34. *Exploring Emerging Technologies in the HPC Co-Design Space*, Barcelona Supercomputing Center, Barcelona, 2015.

35. *Quantitatively Modeling Application Resilience with the Data Vulnerability Factor*, Schloss Dagstuhl, Wadern, 2015.

36. *Exploring Emerging Technologies in the HPC Co-Design Space*, ISC Workshop on Performance Modeling: Methods and Applications, Frankfurt, 2015.

37. *Exploring Emerging Technologies in the HPC Co-design Space*, Qatar Computing Research Institute, Doha, 2014.

38. *Exploring Emerging Technologies in the HPC Co-Design Space*, JST/CREST International Symposium on Post Petascale System Software, Kobe, 2014.

39. *Exploring Emerging Technologies in the HPC Co-Design Space*, Tokyo Institute of Technology, Tokyo, 2014.

40. *Exploring Emerging Technologies in the HPC Co-Design Space*, American Physical Society (APS) March Meeting, Denver, 2014.

41. *Exploring Emerging Technologies in the HPC Co-Design Space*, Clusters, Clouds, and Data for Scientific Computing (CCDSC), Lyon, 2014.

42. *Exploring Emerging Technologies in the HPC Co-design Space*, IBM TJ Watson Research Center, Yorktown, 2014.

43. *Exploring Emerging Technologies in the HPC Co-Design Space*, Institute of Computational Science, Universita della Svizzera italina, Lugano, 2014.

44. *Creating Effective Teams*, DOE CMD-IT National Laboratories Professional Development Workshop for Underrepresented Participants, ORNL, Oak Ridge, 2013.

45. *Exploring Emerging Technologies in the HPC Co-Design Space*, Department of Energy Exhibit @ SC13, Denver, 2013. [more]

46. *Exploring Emerging Technologies in the HPC Co-Design Space*, Tokyo Institute of Technology Exhibit @ SC13, Denver, 2013.

47. *Exploring Emerging Technologies in the HPC Co-Design Space*, NVIDIA GPU Technology Theatre @ SC13, Denver, 2013.

48. *Classifying Soft Error Vulnerabilities in Extreme-Scale Scientific Applications Using a Binary Instrumentation Tool*, SIAM CSE Conference, Boston, 2013.

49. *Intersection of Big Computing and Big Data*, SOS17, Jekyll Island, Georgia, 2013.

50. *Exploring Emerging Technologies in the HPC Co-Design Space*, Swiss Federal Supercomputing Centre, Lugano, 2013.

51. *Vancouver – Unlocking the Potential of Scalable Heterogeneous Systems*, DOE Exascale Research Conference, Arlington, 2012.

52. *Blackcomb Hardware software Co-design for Nonvolatile Memory in Exascale Systems*, SOS16, Santa Barbara, 2012.

53. *The Critical Role of Programming Systems in Future Exascale Systems*, CSE Seminar, University of Michigan, Ann Arbor, 2012.

54. *SHOC and AWE! An update on the Scalable Heterogeneous Computing Benchmark Suite*, Salishan Conference on High-Speed Computing (Random Access session), Lincoln City, OR, 2012.

55. *Aspen: a performance modeling language for codesigning Exascale applications and architectures*, Clusters, Clouds, and Data for Scientific Computing, Lyon, 2012.

56. *Emerging Technologies on the Pathway to Exascale*, Institute for Defense Analysis, Bowie, 2012.

57. *Emerging Technologies on the Pathway to Exascale Computing*, USC-DOE conference on Materials for Energy Applications - Experiment, Modeling and Simulations, Los Angeles, 2011.

58. *Programming Heterogeneous Systems Productively*, SOS15, Engelberg, Switzerland, 2011.

59. *Transformational Computational Science: Past, Present, and Future*, Convocation for Georgia Tech School of Computational Science and Engineering, Atlanta, 2011.

60. *The Under-appreciated Exascale Challenge: Memory and Storage*, International Supercomputing Conference, Hamburg, 2011.

61. *Emerging Architectures on the Pathway to Exascale Computing*, Institute for Nuclear Theory: Extreme Computing and its Applications, Seattle, 2011.

62. *NSF Keeneland Project*, NCAR Workshop on Programming weather, climate, and earth-system models on heterogeneous multi-core platforms, Boulder, 2011.

63. *Toward Exascale Computational Science with Graphics Processors*, Tokyo Institute of Technology, Tokyo, 2009.

64. *HPC Interconnection Networks*, SOS13, Hilton Head, 2009.

65. *HPC Interconnection Networks – The Key to Exascale Computing*, HPC 2008 – Advanced Research workshop on High Performance Computing, Cetraro, Italy, 2008.

66. *Tightly-coupled heterogeneous computing – deux ex machina*, DOE High Performance Computer Science Week, Denver, 2008.

67. *Performance Modeling and Analysis on Multi-Paradigm Computing Systems Challenges and Opportunities*, Dagstuhl Seminar 07341: Code Instrumentation and Modeling for Parallel Performance Analysis, Wadern, 2007.

68. *The Future of Performance Engineering in HPC*, DOE Advanced Scientific Computing Advisory Committee (ASCAC), Washington, DC, 2007.

69. *Application Accelerators: dues ex machina?*, Purdue University – Computing Research Institute, West Lafayette, 2007.

70. *Symbolic Modeling Techniques for Symbolic Modeling Techniques for Next Next-Generation Applications and Generation Applications and Architectures*, DARPA High Productivity Computing Systems Workshop, Denver, 2006.

71. *Application Accelerators: dues ex machina?*, Clusters and Computational Grids for Scientific Computing, Asheville, 2006.

72. *Overview of Future Technologies at ORNL*, Atomic Weapons Establishment (AWE) seminar, Aldermaston, 2005.

73. *Performance Analysis of Ultrascale Systems Using Performance Counters*, SIAM Conf. Parallel Processing for Scientific Computing, San Francisco, 2004.

74. *Performance Modeling and Analysis: Striking a Balance between Resolution and Insight*, Intel Clustering Workshop, Toronto, 2004.

75. *A Progress Report on the Cray X1 Evaluation at the Center for Computational Sciences at ORNL*, Cray Users Group (CUG) Meeting, Knoxville, TN, 2004.

76. *Performance Analysis and Modeling of High End Computing Systems*, Zentrum für Hochleistungsrechnen, Technische Universität Dresden, Dresden, Germany, 2003.

77. *Performance modeling and analysis: striking a balance between resolution and insight*, SC2003 APART Workshop, Phoenix, 2003.

78. *Performance Analysis and Modeling of Terascale Applications: Striking a Balance between Resolution and Insight*, NSF NPACI All Hands Meeting, San Diego, 2003.

79. *Performance Analysis and Modeling of High End Computing Systems*, Performance Evaluation Research Center (PERC) Workshop on Benchmarking, Performance Analysis and Modeling, Berkeley, CA, 2003.

80. *ASCI Experiences: ASCI Tools and Opportunities to Leverage DOE Infrastructure*, NSF NPACI All Hands Meeting, San Diego, 2003.

81. *Performance Analysis and Modeling of High End Computing Systems*, ParCo Symposium on Performance Analysis and Modeling, Dresden, Germany, 2003.

82. *Performance Analysis and Modeling of High End Computing Systems*, University of Rochester Computer Science Colloquium, Rochester, NY, 2003.

83. *Performance Analysis and Modeling of High End Computing Systems*, HPC User Forum, Princeton, 2003.

84. *Performance Analysis and Modeling of High End Computing Systems*, University of Tennessee Computer Science Department, Knoxville, TN, 2003.

85. *Performance Analysis and Modeling of High End Computing Systems*, Oak Ridge National Laboratory, Oak Ridge, 2003.

86. *Scalable performance analysis and prediction techniques for terascale computing*, University of Texas, Austin, Austin, 2002.

87. *Characterization and performance modeling of scalable scientific applications*, Lawrence Berkeley National Laboratory, Berkeley, CA, 2002.

88. *Scalable performance analysis and prediction techniques for terascale computing*, DARPA HPCS PI Meeting, Phoenix, 2002.

89. *Automated techniques for terascale performance analysis*, Lawrence Berkeley National Laboratory, Berkeley, CA, 2001.

90. *A performance data explosion: striking an appropriate balance between resolution and insight*, LACSI Performance Tools Workshop, Santa Fe, 2001.

91. *Techniques for high performance computational steering*, Microsoft Research, Seattle, 1999.

92. *Techniques for high performance computational steering*, University of New Mexico, Albuquerque, 1999.

93. *Techniques for high performance computational steering*, Lawrence Livermore National Laboratory, Berkeley, CA, 1999.

94. *Techniques for high performance computational steering*, University of Arizona, Tucson, 1999.

95. *Techniques for high performance computational steering*, University of Minnesota, Twin Cities, 1999.

6.4 Contributed

1. *Exploring Extreme Heterogeneity with Analytical Modeling*, SIAM PP18 MS111, Tokyo, 2018.

2. *Performance Engineering of Emerging Memory Systems* , SIAM PP18 MS41, Tokyo, 2018.

3. *CLACC: OpenACC for LLVM*, EuroLLVM, Bristol, 2018.

4. *Understanding Portability of a High-Level Programming Model on Diverse HPC Architectures*, DOE Centers of Excellence Performance Portability Meeting, Glendale, AZ, 2016.

5. *Programming Nonvolatile Memory Systems*, Salishan Conference on High Speed Computing (Random Access Talk), Gleneden Beach, Oregon, 2016.

6. *Exploring Emerging Technologies in the HPC Co-Design Space*, Exascale Application and Software Conference, Edinburgh, 2015.

6.5 Panel

1. *Panel: Productivity*, NITRD Workshop on Software in the Era of Extreme Heterogeneity, Virtual, 2020.

2. *Reconfigurable Computing in HPC: Success Stories Today and Future?*, SC19, Denver, 2019.

3. *BoF: Reconfigurable/FPGA Clusters for High Performance Computing*, SC19, Denver, 2019.

4. *BoF: Advanced Architecture Testbeds: Community Resources for Enhanced HPC Research*, SC19, Denver, 2019.

5. *Software and Hardware Support for Programming Heterogeneous Memory*, SC19 MCHPC Workshop, Denver, 2019.

6. *Preparing for Extreme Heterogeneity in HPC*, NITRD Future Computing Community of Interest Meeting, Washington, DC, 2019.

7. *Reconf500: Can we join the 500 club?*, 3rd International Workshop on FPGA for HPC (IWFH), Tokyo, 2018.

8. *Computing in the Post-Moore Era*, SIAM PP18, Tokyo, 2018.

9. *Memory Systems Problems and Solutions*, MEMSYS, Washington, DC, 2018.

10. *Benchmarking Scientific Reconfigurable / FPGA Computing*, SC18, Dallas, 2018.

11. *Volunteer Opportunities for SC Conference Planning*, SC18, Dallas, 2018.

12. *Advanced Architecture Testbeds: A Catalyst for Co-design Collaborations*, SC18, Dallas, 2018.

13. *Revisiting the 2008 ExaScale Computing Study and Venturing Predictions for 2028*, SC18, Dallas, 2018.

14. *Post Moore Supercomputing*, SC17, Denver, 2017.

15. *Post Moore's Era Supercomputing in 20 Years (PMES) Panel*, SC16 Panel, Salt Lake City, 2016.

16. *Towards Exascale Memory Systems*, 17th SIAM Conference on Parallel Processing for Scientific Computing, Paris, 2016.

17. *Parallel Programming*, ISC, Frankfurt, 2016.
18. *Opportunities for HPC based on OpenPOWER Technologies*, International Workshop on OpenPOWER for HPC (IWOPH'16) @ ISC16, Frankfurt, 2016.
19. *Post Moores Era Supercomputing*, SC16, Salt Lake City, 2016.
20. *HPC and Big Data Convergence*, 2nd International HPC Forum, Tianjin, 2015.
21. *Working with Funding Agencies*, NSF Academic Workshop in Computing for Underrepresented Participants, Atlanta, 2012.
22. *Exascaling your Applications*, NVIDIA GTC 2012, San Jose, 2012.
23. *DARPA UHPC: Echelon and CHASM*, HPCC, Newport, RI, 2011.
24. *A Multidisciplinary Approach to Resilience*, International Supercomputing Conference, Hamburg, Germany, 2009.
25. *Is PGAS right for heterogeneous computing*, PGAS Conference, Ashburn, VA, 2009.
26. *Programming Accelerator Clusters — Where Do We Go From Here?*, Workshop on Parallel Programming on Accelerator Clusters (PPAC), New Orleans, 2009.
27. *Concurrency-Related Challenges*, Advanced Computing Systems Research Program, Annapolis, 2009.
28. *Past, Present, and Future of Parallel Processing – ICPP 35th Anniversary Panel*, International Conf. Parallel Processing, Columbus, 2006.
29. *How video games and multimedia will affect the HPC industry?*, International Conf. Parallel Processing, Oslo, 2005.
30. *Future Directions for HPC*, SOS9 Workshop, Davos, 2005.
31. *High Performance Computing: Successes, Failures, and Future Directions*, ACS/IEEE International Conference on Computer Systems and Applications, Cairo, 2005.
32. *A Science-Based Path to Petascale Computing*, Intel HPC Roundtable, Portland, OR, 2004.
33. *Design and evaluation of future computing architectures*, Fall Creek Falls Computing workshop, Fall Creek, TN, 2004.
34. *HPCchallenge Benchmarks: An Expanded View of High End Computer (HEC) Performance*, SC2004 High Performance Computing, Networking, and Storage, Pittsburgh, 2004.
35. *OS and System Software for Ultrascale Architectures*, SOS8 Workshop, Charleston, 2004.

6.6 Tutorial

1. *Scalable Heterogeneous Computing on GPU Clusters*, SC12, Salt Lake City, 2012.
2. *M12: Scalable Heterogeneous Computing on GPU Clusters*, SC11, Seattle, 2011.
3. *NSF Keeneland GPU Tutorial and Workshop*, Georgia Institute of Technology, Atlanta, 2011.

7 Software Products

1. IRIS: Intelligent Runtime Systems for Heterogeneous Computing. https://code.ornl.gov/eck/brisbane-rts
2. Siena: exploring the design space of heterogeneous memory systems. https://github.com/ORNL/siena-mem
3. Papyrus: Parallel Aggregate Persistent Storage. https://code.ornl.gov/eck/papyrus
4. DESTINY: a 3D design-space exploration tool for SRAM, eDRAM and non-volatile memory. https://code.ornl.gov/3d_cache_modeling_tool/destiny
5. Open Accelerator Research Compiler (OpenARC). http://ft.ornl.gov/research/openarc
6. SHOC – Scalable HeterOgeneous Computing Benchmark Suite. https://github.com/vetter/shoc
7. mpiP – Lightweight, scalable MPI profiling tool. Verified to over 200k processors. http://www.llnl.gov/casc/mpip
8. Sequoia – Tool for application performance modeling and characterization.
9. Umpire – Tool for verifying the correctness of MPI applications dynamically at runtime. Inspired Intel's Message Checker™ in the Intel Cluster Toolkit™.
10. Photon – Novel implementation of MPI that supports statistical message sampling for performance analysis.
11. Performance assertions – Annotation and runtime toolkit for explicit assertion of performance expectations in HEC applications.

8 Patents

1. *Performance analysis of distributed applications using automatic classification of communication inefficiencies*. US Patent 6,850,920. Application awarded February 2005.

9 Professional Activities and Service

9.1 Leadership

1. 2020, SRC Decadal Plan for Semiconductors, Executive Committee
2. 2020, NITRD Workshop on Software in the Era of Extreme Heterogeneity, Advisor
3. 2018, HPC Asia 2018: International Conference on High Performance Computing in Asia Pacific Region, Steering Committee
4. 2018, ASCR Workshop on Extreme Heterogeneity in High Performance Computing, Chair
5. 2018, Third International Workshop on Post Moores Supercomputing, Co-chair
6. 2017, Workshop on HPC Computing in a Post Moore's Law World (HCPM), Co-chair
7. 2017, IEEE Micro: Special Issue on Architectures for the Post-Moore Era, Co-Guest Editor
8. 2017, ISC, Research Posters, Chair
9. 2017, Dagstuhl Seminar 17431: Performance Portability in Extreme Scale Computing: Metrics, Challenges, Solutions, Co-organizer
10. 2017, SC17, Birds-of-a-Feather Chair
11. 2017, Second International Workshop on Post Moores Supercomputing, Co-chair
12. 2016, International Supercomputing Conference (ISC), Steering Committee
13. 2016, ACM Journal on Emerging Technologies in Computing (JETC), Associate Editor
14. 2016, First International Workshop on Post Moores Supercomputing, Co-chair
15. 2016, Exascale Requirements Review for Advanced Scientific Computing Research (ASCR), Co-chair
16. 2015, SC15, Technical Program Chair
17. 2015, ACM Gordon Bell Prize, Deputy Chair
18. 2015, SC, Steering Committee
19. 2014, SC14, Technical Program Deputy Chair
20. 2014, U. Tsukuba (Japan), Center for Computational Sciences, Chair of External Review Board
21. 2014, SC14, Emerging Technologies Co-chair
22. 2014, Workshop on accelerator programming using directives (WACCPD), Steering Committee
23. 2012, SC12, Technical Papers Co-chair
24. 2011, ACM SIGPLAN Workshop on Memory Systems Performance and Correctness (MSPC), General Chair
25. 2011, MTAAP 2011, Co-chair
26. 2010, IPDPS 2010 - IEEE International Parallel and Distributed Processing Symposium, Vice chair for applications
27. 2010, MTAAP 2010 - Workshop on Multithreaded Architectures and Applications, Co-chair
28. 2010, ACM SIGPLAN Workshop on Memory Systems Performance and Correctness, Steering Committee

29. 2010, Cross-cutting Technologies for Computing at the Exascale Workshop, Programming Models Theme Leader
30. 2008, DOE Institute for Advanced Architectures and Algorithms - Interconnections Networks Workshop, Co-chair
31. 2007, DOE Software Development Tools for Petascale Computing Workshop, General Chair
32. 2007, SC07, Disruptive Technologies Chair
33. 2007, International Conference on Parallel Processing (ICPP), Vice-Chair for Software Systems
34. 2007, IEEE Transactions on Computers, Associate Editor
35. 2006, Euro-Par 2006, Vice Chair for Performance
36. 2005, SC05, Technical Papers Area Chair for Performance, Tutorials Committee
37. 2004, Euro-Par 2004, Vice Chair for Performance Evaluation
38. 2004, International Conference on Parallel Processing (ICPP), Vice-Chair for Performance Evaluation
39. 2002, Los Alamos Computer Science Institute (LACSI) Symposium Workshop on Performance Analysis with R. Barrett and A. Malony, Co-chair
40. 2001, Los Alamos Computer Science Institute (LACSI) Symposium Workshop on Performance Analysis with R. Barrett and A. Malony, Co-chair

9.2 Regular

1. 2021, DOE ASCR Workshop on Reimagining Codesign, Organizing Committee
2. 2021, PPOPP, Program Committee
3. 2020, SC20 Invited Speakers, Program Committee
4. 2019, DOE ASCR 40 Year Highlights, Organizing Committee
5. 2019, IEEE Conference on Rebooting Computing (ICRC), Program Committee
6. 2019, PPOPP, Program Committee
7. 2019, IEEE Computer Society Fellow Evaluation Committee, Program Committee
8. 2019, DOE Artificial Intelligence for Science Town Halls, Technical Topic Leader
9. 2018, PPOPP, Program Committee
10. 2018, SIAM Parallel Processing Conference, Organizing Committee
11. 2018, IEEE Computer Society Fellow Evaluation Committee, Program Committee
12. 2018, Re-Emergence of Vector Architectures (REV-A), Program Committee
13. 2017, International Parallel and Distributed Processing Symposium (IPDPS), Program Committee
14. 2017, PPOPP, Program Committee
15. 2017, International Symposium on Performance Analysis of Systems and Software (ISPASS), Program Committee
16. 2017, High Performance Extreme Computing Conference (HPEC), Program Committee
17. 2017, Re-Emergence of Vector Architectures (REV-A), Program Committee

18. 2017, 3rd Workshop on Heterogeneous High-performance Reconfigurable Computing , Program Committee
19. 2017, MCHPC2017: Workshop on Memory Centric Programming for HPC, Program Committee
20. 2016, High Performance Extreme Computing Conference, Program Committee
21. 2016, MODSIM, Organizing Committee
22. 2016, PMMA16, Organizing Committee
23. 2016, ISC, Program Committee
24. 2016, ICS, Program Committee
25. 2016, SC16, Program Committee, Technical Papers
26. 2015, MODSIM, Organizing Committee
27. 2015, ASPLOS, Program Committee
28. 2015, High Performance Extreme Computing Conference, Program Committee
29. 2014, High Performance Extreme Computing Conference, Program Committee
30. 2014, MTAAP, Program Committee
31. 2014, ICS, Program Committee
32. 2014, PPOPP, Program Committee
33. 2014, DOE ASCAC Workforce Subcommittee, Advisory/Review Committee
34. 2014, DOE MODSIM 2014, Organizing Committee
35. 2013, MTAAP, Program Committee
36. 2013, International Parallel and Distributed Processing Symposium (IPDPS), Program Committee
37. 2013, GPGPU6, Program Committee
38. 2013, SC13, Panels Committee; Gordon Bell Committee; Emerging Technologies Committee
39. 2013, HiPC, Program Committee
40. 2013, DOE MODSIM 2013, Organizing Committee
41. 2013, MSPC 2013, ACM SIGPLAN Workshop on Memory Systems Performance and Correctness, Program Committee
42. 2013, High Performance Extreme Computing Conference, Program Committee
43. 2012, ACM InPar, Program Committee
44. 2012, PPOPP, Program Committee
45. 2012, ACM ICS, Program Committee
46. 2012, ISC 2012, Program Committee, Tutorials
47. 2011, ACM Workshop on General Purpose Processing on Graphics Processing Units, Program Committee
48. 2011, SC11, Program Committee, Tutorials

49. 2011, SciDAC 2011, Organizing Committee
50. 2011, ACM Workshop on Emerging Applications and Many-Core Architectures (EAMA), Organizing Committee
51. 2011, First International Workshop on Characterizing Applications for Heterogeneous Exascale Systems, Program Committee
52. 2010, SIAM Parallel Processing Conference, Organizing Committee
53. 2010, SC10, Program Committee, Tech Papers
54. 2009, Workshop on Parallel Programming on Accelerator Clusters (PPAC), Program Committee
55. 2009, SC09, Program Committee, Tech Papers
56. 2009, DOE Institute for Advanced Architectures and Algorithms - Workshop on HPC Architectural Simulation (HPCAS-1), Organizing Committee
57. 2009, MTAAP, Program Committee
58. 2008, IEEE International Symposium on Workload Characterizations, Program Committee
59. 2008, Principles and Practice of Parallel Programming (PPOPP), Program Committee
60. 2008, Workshop on Multithreaded Architectures and Applications (MTAAP), Program Committee
61. 2008, SC08, Tutorials Committee, Disruptive Technology Committee
62. 2008, SIAM Parallel Processing Conference, Mini-symposium Organizer
63. 2008, International Exascale Software Project, Organizing Committee
64. 2007, Institute for Advanced Architectures Workshop on Memory Opportunities for High Performance Computing, Program Committee
65. 2007, DOE/DOD Workshop on Emerging High Performance Architectures and Applications, Organizing Committee
66. 2007, IEEE International Symposium on Workload Characterizations, Program Committee
67. 2007, ACM Computing Frontiers, Program Committee
68. 2006, SC06, Exotic Technologies Initiative Committee, Technical Papers Committee
69. 2006, International Parallel and Distributed Processing Symposium (IPDPS), Program Committee
70. 2005, Parallel Computing (ParCo), Program Committee
71. 2005, ACM SIGMETRICS, Program Committee
72. 2005, High Performance Distributed Computing (HPDC), Program Committee
73. 2005, LCI International Conference on Linux Clusters: The HPC Revolution, Program Committee
74. 2005, Workshop on System Management Tools for Large-Scale Parallel Systems, Program Committee
75. 2005, International Parallel and Distributed Processing Symposium (IPDPS), Program Committee
76. 2004, International Conference for High-Performance Computing, Program Committee
77. 2004, Seventh Workshop on Languages, Compilers, and Runtime Support for Scalable Systems (LCR), Program Committee

78. 2003, International Parallel and Distributed Processing Symposium (IPDPS), Program Committee
79. 2003, International Conference on Computational Science, Workshop on Performance Evaluation, Modeling and Analysis of Scientific Applications on Large-Scale Systems, Program Committee
80. 2003, SC03, Program Committee
81. 2003, Parallel Computing (ParCo), Program Committee
82. 2003, Los Alamos Computer Science Institute Symp, Program Committee
83. 2003, National Science Foundation, Proposal Review Panelist
84. 2002, Georgia Tech Center for Experimental Research in Computer Systems, Advisory Panel
85. 2002, LCI International Conference on Linux Clusters: The HPC Revolution, Program Committee
86. 2002, Department of Energy NERSC, Computational Review Panel
87. 2002, Department of Energy. Advanced Scientific Computing Research (ASCR) proposals, Proposal Review Panelist
88. 2001, National Science Foundation, Proposal Review Panelist
89. 2001, Science Foundation Ireland, Proposal Review Panelist
90. 2000, Lawrence Livermore National Laboratory University Collaborative Research Program proposals, Proposal Review Panelist
91. 1998, National Science Foundation, Proposal Review Panelist

10 Funding

10.1 Current

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2019	3	DOD	Brisbane: Productive Programming Systems in the Era of Extremely Heterogeneous and Ephemeral Computer Architectures	$2,250	PI
2018	4	DARPA	Cosmic Castle: Enabling Codesign of specialized accelerators	$6,000	PI
2017	3	DOE	Panorama360: Performance data capture and analysis for end-to-end scientific workflows	$3,000	Co-PI
2017	3	DOE	Sawtooth: Investigating solutions to memory and storage challenges	$12,000	PI
2017	3	DOE	RAPIDS: SciDAC4 Computer Science and Data Institute	$18,000	Senior
2016	3	DOE	Exascale Computing Project (L3 Manager for Compilers and Tools)	$n/a	Manager
2019	4	DOE	PROTEAS-TUNE: Programming Systems for Emerging Architectures	$13,200	PI

10.2 Pending

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2020	5	DOE	RAPIDS: SciDAC5 Computer Science and Data Institute	$30,000	Senior

10.3 Completed

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2017	1	DOE	Preparing for Extreme Heterogeneity in High Performance Computing [Workshop]	$40	OC Member
2016	2	LDRD	Tahoe: Designing and programming exascale memory hierarchies	$720	PI
2016	3	DOE	PROTEAS: Programming Systems for Emerging Architectures	$4,500	PI
2014	3	DOE	Panorama: Predictive Modeling and Diagnostic Monitoring of Extreme Science Workflows	$4,500	Co-PI
2014	3	DOE	ARES: Abstract Representations for Extreme-Scale Software Stack	$3,000	Co-PI
2013	3	DOE	Blackcomb2: Hardware-Software Co-design for Non-Volatile Memory in Exascale System	$2,100	PI
2013	3	DOE	Vancouver2: Designing a Next-Generation Software Infrastructure for Productive Heterogeneous Exascale Computing	$2,100	PI
2013	3	DOE	Oxbow: Synthesizing architectural features for extreme-scale co-design applications	$2,250	PI
2013	3	DOE	Hobbes: OS and Runtime Support for Application Composition	$2,500	Senior
2013	3	DOE	Uncertainty Quantification Methodologies for Enabling Extreme-Scale Science	$2,250	Co-PI
2011	5	DOE	Center for Exascale Simulation of Advanced Reactors (CESAR) Co-Design Center	$20,000	Co-PI
2011	5	DOE	Exascale Co-Design Center for Materials in Extreme Environments (ExMatEx)	$20,000	Co-PI
2010	3	DOE	Blackcomb: Hardware-Software Co-design for Non-Volatile Memory in Exascale System	$3,000	PI
2010	3	DOE	Vancouver: Designing a Next-Generation Software Infrastructure for Productive Heterogeneous Exascale Computing	$2,600	PI
2010	6	NVIDIA	NVIDIA CUDA Center of Excellence (industry grant)	$700	PI
2010	1	Reservoir Labs	Advance Static and Dynamic Mapping of HPC Applications on Petascale Systems	$60	Co-PI
2010	2	DARPA	Echelon: Extreme-scale Compute Hierarchies with Efficient Locality-Optimized Nodes	$25,000	Co-PI

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2010	2	DARPA	CHASM: Challenge Applications and Scalable Metrics for Ubiquitous High Performance Computing	$7,500	Co-PI
2010	1.5	DOE	SciDAC-e: Enhancing Productivity of Materials Discovery Computations for Solar Fuels and Next Generation Photovoltaics	$110	Co-PI
2009	5	NSF	Keeneland: National Institute for Experimental Computing / 0910735 / Vetter	$12,000	PI
2009	4	NSF	CI-ADDO-NEW: MANIFOLD- A Scalable Simulation Infrastructure for Future Many Core Systems	$1,300	Senior
2009	3	DOE	MOGO: Model Oriented Global Optimization of Petascale Scientific Applications	$1,500	PI
2008	3	DARPA	Blackjack: Evaluation of Architecture-Aware Compiler Environments	$4,500	Co-PI
2008	4	DARPA	Productivity Evaluation of HPCS Architectures	$3,600	PI
2008	1	Advanced Micro Devices	Geneva: Performance Prediction for Tightly Coupled Heterogeneous Processors	$50	PI
2007	3	NSF	NSF CRI: Development of a Research Infrastructure for the Multithreaded Computing Community Using the Cray Eldorado Platform	$80	
2006	2	ORNL LDRD	An Evolutionary Framework for Porting Applications to Petascale Platforms	$500	
2006	5	DOE	SciDAC Performance Engineering Research Institute	$15,000	Co-PI
2006	5	DOE	SciDAC Scientific Data Management Center	$16,500	Senior
2005	2	ORNL LDRD	Automated Performance Tools for Petascale Systems with Lightweight Compute Node Kernels	$550	
2004	4	National Security Agency	Petascale Single System Image	$1,500	
2004	3	DOE	Modular Linux and Adaptive Runtime support for HEC OS/R Research	$1,500	
2004	2	ORNL LDRD	Exploring Alternative Technologies for Next-Generation Leadership Class Computing	$	
2004	4	DOE	HPCS PetaScale Execution Time Evaluation	$900	
2004	3	DOD HPC-MOD	Specialized Performance Analysis and Modeling of Department of Defense High Performance Computing Modernization Program Applications	$540	
2004	2	DOE	Performance Evaluation Research Center	$	

10.4 Not funded

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2019	3	DOE	Co-design of in situ Scientific Machine learning Integrated Components	$2,000	Senior
2018	1	NSF	Collaborative: SI2-S2I2 Conceptualization: Software Institute for Robust Science: Reproducibility, Automation and Trust in Software for Science	$	Co-PI
2017	3	DOE	Quantum Sampling Algorithms for Scientific Applications of Machine Learning	$4,500	Senior
2016	3	DOE	Automation for Robust Exascale Workflow Applications	$3,750	Co-PI
2016	3	DOE	Montreux: Performance Characterization, Prediction, and Measurement of ECP Applications	$2,250	PI
2015	4	DARPA	CATALYST (Building Resource Adaptive Software Systems for Heterogeneous, Scalable Computing Systems)	$8,500	PI
2015	3	DOE	Ashland: Globally Optimized Dynamic Resource Management for Distributed Scientific Workflows	$4,500	PI
2015	3	DOE	Silverburn: Measuring, Understanding, and Predicting Behaviors of Emerging Extreme-scale Storage Paradigms	$2,420	PI
2015	3	DOE	Whistler: Designing Resilient Next-Generation Memory Systems	$2,100	Co-PI
2015	3	DOE	Resilience Building Blocks	$3,750	Co-PI
2014	5	NSF	FG Cloud	$10,000	Co-PI
1	3	DOE	IDEAS: Interoperable Design of Extreme-scale Application Software	$12,000	Senior
2013	3	DOE	ExaMover: Adaptive Network Transport and Unified I/O Management Framework for Moving Exabyte Data over WAN	$1,800	PI

Start	Duration (Years)	Sponsor	Title	Total Award ($K)	Role
2013	3	NSF	ABI development: Improving Protein Domain Modeling using Ultrascale Computing	$1,100	Co-PI
2013	5	DOE	E3-S: Exascale End-to-End Solvers for PDE-Based simulation and optimization problems	$4,500	Co-PI
2012	2	ARO	Pinehurst	$1,500	PI
2012	5	DOE	Towards Predictive Exascale Simulations of Materials Interfaces in Extreme Environments	$1,350	Co-PI
2011	5	DOE	SciDAC3: IDEAL: Institute for Data movement and Energy Optimization in Applications (IDEAL)	$3,000	Co-PI
2011	3	NSF	G8: Tempo	$600	Co-PI
2011	3	NSF	G8: XMG	$600	Co-PI
2010	3	NSF	FastReduce: Infrastructure for end-to-end exascale fast multipole methods	$500	Co-PI
2010	3	DOE	Whistler: Integrated Interconnection Networks for Exascale Computing	$100	PI
2010	3	DOE	Breckenridge: GPU Memory Hierarchies for Exascale Systems	$3,600	PI
2010	3	DOE	An Open Integrated Software Stack for Extreme Scale Computing	$15,000	
2010	3	DOE	Large Data Systems – Computational Data Flows for Exascale Machines	$2,700	
2010	5	DOE	Fusion Reactor Exascale Simulation Co-design Center (FRESCO)	$40,000	Co-PI
2010	5	DOE	Combustion Exascale Co-Design Center	$40,000	Co-PI
2010	5	DOE	Exascale Center for Electrochemical Energy storage Devices (ExCEED)	$40,000	Co-PI
2010	3	NSF	Camden: Accelerating the deployment of the Keeneland ID system for production availability	$3,000	PI
2010	1.5	DOE	SciDAC-e: Accelerating Simulations of Interfacial Transport and Reactivity in Energy Materials	$700	Co-PI
2009	3	NSF	A Scalable Metadata Management Framework for ExaByte File Systems	$1,000	
2009	3	DOE	Designing a Distributed InfiniBand Network for Petascale Science	$1,500	
2009	3	DOE	Exascale Computing of Materials under Extreme Conditions for Energy Technologies	$4,500	
2009	3	DOE	Novel Parallel Numerical Algorithms for Emerging Exascale Computing Platforms	$3,300	
2009	3	NSF	Heterogeneous Virtual Machine: Future Execution Environments for Heterogeneous Many-core Architectures	$1,000	Senior
2009	4	NSF	Manifold – A Scalable Simulation Infrastructure for Future Many Core Systems	$1,679	Senior
2009	3	DOE	Extreme-Scale Correctness Tools	$1,800	
2008	1.5	DOE	IAA Interconnection Networks for Exascale Computing	$1,000	Co-PI
2007	3	DOE	System Software for Tightly Coupled Heterogeneous Computing	$2,070	
2007	3	DOE	Breaking the Collective Wall for Ultra-Scale Parallel IO	$2,099	
2007	3	DOE	Global Storage-Class Memory to Enable Efficient PetaByte Data Accesses for PetaScale Scientific Applications	$2,000	
2007	2	ORNL LDRD	An Integrated Framework to Improve Productivity and Performance on Emerging Processor Architectures	$700	
2007	2	ORNL LDRD	Enabling Efficient Parallel IO for UltraScale Computing	$800	
2007	3	DOE	High Performance Computing Ecosystem Service Oriented Architecture (HPCE SOA)	$1,140	
2006	1.5	NSF	National Innovation Collaboration Ecosystem (NICE) Computing Convergence Study (CCS)	$200	
2006	5	DOE	Center for Scientific Data-In-Transit	$	
2006	5	DOE	Real-Time Data Reduction with Field Programmable Gate Arrays (FPGAs)	$	
2006	5	DOE	Providing Solutions to Critical Software Gaps to Petascale Science	$	
2006	5	DOE	Institute for Enabling High-Performance Applications on Emerging Architectures	$	
2006	5	DOE	Chemistry SAP	$	

11 Teaching and Student Guidance

11.1 Teaching

1. Fall 2007, Georgia Tech CS6230, Graduate-level Introduction to HPC Tools and Applications.
2. Fall 2010, Georgia Tech CS6230, Graduate-level Introduction to HPC Tools and Applications.
3. Spring 2012, Georgia Tech CSE6220, Graduate-level Introduction to HPC.

11.2 Postdoctoral Fellows Supervised

1. Dr. Sadaf Alam, Oak Ridge National Laboratory
2. Dr. Anthony Danalis, Oak Ridge National Laboratory and University of Tennessee
3. Dr. Mehmet Beliveranli, Oak Ridge National Laboratory, 2016–2017
4. Dr. Jungwon Kim, Oak Ridge National Laboratory
5. Dr. Graham Lopez, Oak Ridge National Laboratory
6. Dr. Collin McCurdy, Oak Ridge National Laboratory
7. Dr. Sparsh Mittal, Oak Ridge National Laboratory, 2014–2016
8. Dr. Ivy ”Bo” Peng, Oak Ridge National Laboratory, 2017–2018
9. Dr. Kittisak Sajjapongse, Oak Ridge National Laboratory, 2016
10. Dr. Srinivas Sridharan, Oak Ridge National Laboratory
11. Dr. Dwaipayan Chakraborty, Oak Ridge National Laboratory, 2019–2020.

11.3 M.S. Committee Member

1. Sundaresan Venkatasubramanian, Georgia Institute of Technology, 2009

11.4 Ph.D. Committee Member

1. Srinivas Sridharan, University of Notre Dame
2. Wenbin Zhu, University of New Mexico, 2008

11.5 M.S. Students Supervised

1. Viraj Rameshrao Paropkari, Georgia Institute of Technology, 2007
2. Kyle Spafford, Georgia Institute of Technology, 2008-9

11.6 Student Internships Supervised

1. Sreesa Akella, 2005, University of South Carolina-Columbia
2. Rshiraj Bheda, 2015, 2016, Georgia Tech
3. Yong Chen, 2010, Illinois Inst. of Technology
4. Hsiang-Yun Cheng, 2014, Penn State University
5. Luis Cordova, 2005, University of South Carolina-Columbia
6. Tristan Cuevas, 2008, Kent State University
7. Anthony Danalis, 2006, University of Delaware
8. Ashley DeFlumere, 2008, Mount Holyoke College
9. Keisuke Fukuda, 2013, 2014, Tokyo Tech
10. Andrew Gearhart, 2008, University of Delaware

11. Seonglyong Gong, 2016, University of Texas, Austin
12. Randy Hassell, 2007, Old Dominion University
13. Tetsuya Hoshino, 2014, Tokyo Tech
14. Matthew Jallouk, 2011, 2012, 2013, University of Tennessee-Chattanooga
15. David LaBissoniere, 2006, East Tennessee State University
16. Joseph Lambert, 2006, Bucknell University
17. Matthew Larsen, 2014, University of Oregon
18. Robert Lim, 2014, University of Oregon
19. Charles Lively, 2008, Texas A&M University
20. Kelly Livingston, 2008, University of Delaware
21. Nicholas Murphy, 2007, University of Washington
22. Viraj Paropkari, 2007, Georgia Institute of Technology
23. David Poliakoff, 2012, University of Oregon
24. Matthew Poremba, 2014, Penn State University
25. Amit Sabne, 2014, Purdue University
26. Putt Sakdhnagool, 2014, Purdue University
27. Samuel Schlegel, 2015, Florida Atlantic University
28. Kyle Spafford, 2008, Georgia Institute of Technology
29. Srinivas Sridharan, 2008, University of Notre Dame
30. Junqing Sun, 2007-8, University of Tennessee-Knoxville
31. Philip Taffet, 2014, Rice University
32. Nirmal Thacker, 2008, Georgia Institute of Technology
33. Mariam Umar, 2016, Virginia Tech
34. Karthik Vijayakumar, 2009, North Carolina State University
35. Rujia Wang, 2016, University of Pittsburgh
36. Bo Wu, 2013, College of William and Mary
37. Panruo Wu, 2013, University of California – Riverside
38. Jake Wynne, 2015, University of Tennessee
39. Li Yu, 2014, Illinois Institute of Technology
40. Brandon Kammerdiener, 2019, University of Tennessee
41. David Pruitt, 2019, University of Texas, El Paso
42. Jacob Lambert, 2018-2019, University of Oregon
43. Pak Markthub, 2017-18, Tokyo Institute of Technology

12 Personal

1. Citizenship: USA.

2. Security clearances: Available upon request.

13 References

- See attached reference list.